

A Generalized Approach to Combining Similarity and Eigenvector Analysis of Networks*

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Abstract

This paper presents a generalized approach to combining two key ideas in social network analysis: pairwise node similarity (based on local neighborhoods) and spectral ranking using feedback centralities (such as Bonacich’s eigenvector scores). Building on the work of Alvarez-Socorro et al. (2015), who proposed weighing network edges by the dissimilarity between connected nodes to favor brokerage (connecting to well-connected, dissimilar others), the proposed generalization allows analysts to specify either dissimilarity or similarity as the preferred criterion. Weighing by similarity is argued to be more consistent with the idea of prominence as membership in a centralized core of well-connected nodes (closure). The generalized approach introduces parameters (α and β) to combine these two strategies, yielding a compromise ranking that accounts for both brokerage and closure. An analysis of the classic Florentine Families network demonstrates that the Medici family specializes in brokerage, ranking highest when a dissimilarity weight is used. In contrast, the Peruzzi family emerges as the most central when an approach that rewards similarity/closure is used.

Introduction

When analyzing network data, a natural first task is to get a sense of which nodes in the network are more “important” or “prominent” given some conception of a process unfolding within the structural configuration (Borgatti 2005). This exposes the analyst to the wide world of centrality analysis and the various measures and indices contained therein (Borgatti and Everett 2006). Another follow-up task when analyzing networks is to examine relations between nodes based on structural information, which refers to either the proximity or differences in the connectivity structure of a pair of nodes in the graph, opening up the analysts to the equally wide world of similarity and distance measures for nodes in graphs (Fouss et al. 2005; Leicht et al. 2006).

Perhaps because they are seen as involving distinct goals (e.g., centrality analysis is for ranking nodes, while similarity/distance analysis is for clustering nodes), the *linkages* between these two types of tasks have not been remarked as much. This methodological note builds on a recent effort to establish one such linkage (Alvarez-Socorro et al. 2015), asking how we can use pairwise structural information from nodes in a graph to augment the usual centrality analysis ranking, with a focus on the so-called “feedback” centralities based on computing the leading eigenvector of a node proximity matrix (usually taken to be the adjacency matrix).

The basic idea is that we can use the output of some kind of similarity computation (which usually also entails a distance between nodes) as a *weight* on the edges in the graph, so as to modify the usual feedback centrality computation towards considering some connections as more “important” than others in determining the centrality of the focal node. In this way, the similarity/distance analysis is used as input to “tweak” the computation of centrality-based ranking, based on a (theoretically motivated) intuition that edges with certain features (e.g., linking the focal node to either similar or dissimilar others) should count more, thus relaxing the usual assumption that all edges (unless supplemented with some exogenous information) are exchangeable. This offers a more flexible approach to centrality scoring in networks that requires only endogenous information from the network, based on the connectivity structure encoded in the usual adjacency matrix.

The rest of the paper is organized as follows. In the next section, I review the standard “feedback” based centrality due to Bonacich (1972), and then go on to consider a recent proposal to augment this approach using pairwise *dissimilarity* scores on the edges of the graph due to (Alvarez-Socorro et al. 2015). I then propose a generalized approach that allows the analyst to weigh edges by *either* the similarity or the dissimilarity (or to combine both considerations at once), and use the oft-considered example of the Florentine Families network (Padgett and Ansell 1993) to show how we can obtain substantively distinct node rankings based on whether we weigh edge importance by the similarity or dissimilarity when obtaining the centrality scores. I close by pointing to some future lines of further research and application.

Eigenvector Centrality

The standard approach to computing “feedback”-based indices of node position in networks is due to the classic work of Bonacich (1972), subsequently developed in Bonacich (1987). Here, the idea of feedback reduces to the more specific notion that a node’s ranking in the resulting index—which can be interpreted as a “centrality” measure (Borgatti and Everett 2006)—is higher if it is connected to others who are themselves well-connected. Another way of saying this is that the resulting centrality score of a given node should be *proportional* to the *sum* of the same score for each one of their neighbors. Thus, if the (undirected) adjacency relations in the network are represented in the (symmetric) matrix $\mathbf{A}$ with $a_{ij} = 1$ if nodes i and j are connected in the underlying graph, then the vector of standard Bonacich scores $\mathbf{s}$ can be obtained via the solution to the linear system:

$$\lambda_1 \mathbf{s} = \mathbf{A} \mathbf{s} \tag{1}$$

Where λ_1 is the leading (largest) eigenvalue of the adjacency matrix, also referred to as $\rho(\mathbf{A})$, or the “spectral radius” of the adjacency matrix. If λ_1 exists, then its associated $\mathbf{s}$ eigenvector also exists, and it is guaranteed (via the famous Perron-Frobenius theorem) to be composed of real non-negative scores, which can be used to induce a ranking on the nodes of the network (Vigna 2016).

Combining Similarity and Eigenvector Ranking

In a recent contribution, Alvarez-Socorro et al. (2015) propose to merge the Bonacich approach to feedback centrality scoring with the analysis of vertex similarity in networks (Leicht et al. 2006). Their basic idea is simple, proposing that we substitute the adjacency matrix in Equation 1 with a different matrix $\mathbf{W}$ which is defined as:

$$\mathbf{W} = \mathbf{A} \odot (1 - \mathbf{S}) \tag{2}$$

Where $\mathbf{S}$ is a matrix of (local) vertex *similarities* (making $1 - \mathbf{S}$ a dissimilarity or distance matrix), and the $\odot$ operation refers to the element-wise “dot” product. In essence, what Alvarez-Socorro et al. (2015) are doing is attaching the local vertex *dissimilarity* score—bound to fall in the zero to one interval—as a *weight* on each edge in the network, and then computing the standard Bonacich scores from the dissimilarity weighted adjacency matrix to rank nodes in the network.

Note that this approach sets the dissimilarity scores of non-adjacent nodes to zero, making the results different from the spectral decomposition of the raw distance matrix $1 - \mathbf{S}$. Note also that if the computation of $\mathbf{S}$ assigns $s_{ij} = 1.0$ to vertex pairs with maximum structural similarity (e.g., structurally equivalent pairs), then the Alvarez-Socorro et al. (2015) approach will set the edge weights between structurally similar

nodes to $w_{ij} = 0$ in $\mathbf{W}$ thus dropping them from consideration in determining the centrality of the respective nodes.

This leads to the modified linear system:

$$\lambda_1 \mathbf{s} = \mathbf{W} \mathbf{s} \quad (3)$$

There are many options from which to construct $\mathbf{S}$, but an attractive one is the Jaccard similarity s^J , because its complement $1 - s^J$ returns a true metric distance (respecting the triangle inequality). In terms of the adjacency matrix, we can define the Jaccard similarity matrix as:

$$\mathbf{S} = \frac{\mathbf{P}}{\mathbf{P} + \mathbf{Q} + \mathbf{Q}^T} \quad (4)$$

With:

$$\mathbf{P} = \mathbf{A}^2 \quad (5a)$$

$$\mathbf{Q} = \mathbf{A}(1 - \mathbf{A}) \quad (5b)$$

Why would we want to modify the usual eigenvector centrality computation in the way suggested by Equation 3? As Alvarez-Socorro et al. (2015) note, their approach is motivated by the idea that being connected to well-connected others who are also *dissimilar* from you (because they are not linked to your neighbors) should matter more in determining node ranking than being connected to well-connected others who are also connected to your neighbors, since dissimilar others provide exclusive points of access to contacts who cannot be reached directly while similar others provide redundant access to contacts that can be reached directly.

Thus, theoretically, the Alvarez-Socorro et al. (2015) approach merges the “feedback” reasoning animating the Bonacich prestige ranking approach with the “brokerage” reasoning of Burt’s (1995) theory of structural holes. Thus, the “value” of a connection to someone with whom you don’t share any neighbors is seen as higher than a connection to someone with whom you share lots of neighbors in determining your own centrality. This is, of course, not a completely unreasonable theoretical assumption to make, but as we will see in a bit, it is not the *only* justifiable one we can make.

Algorithm 1 presents pseudocode for a simple algorithm that implements Alvarez-Socorro et al.’s eigenvector-based node-scoring approach.

A Generalized Approach

As we just noted, the Alvarez-Socorro et al. (2015) approach to eigenvector scoring assumes that we want to characterize nodes as most central when their connectivity strategy combines a logic of proximity to the well-connected with a logic of brokerage between disconnected others. However, in some applications, the most central nodes may

Algorithm 1 Alvarez-Socorro’s et al.’s (2015) Dissimilarity Scoring

Require: $\mathbf{A}$	$\triangleright n \times n$ symmetric adjacency matrix
1: $\mathbf{P} \leftarrow \mathbf{A}^2$	$\triangleright$ Common neighbors matrix
2: $\mathbf{Q} \leftarrow \mathbf{A}(1 - \mathbf{A})$	$\triangleright$ Non-shared neighbors matrix
3: $\mathbf{S} \leftarrow \mathbf{P} \div (\mathbf{P} + \mathbf{Q} + \mathbf{Q}^T)$	$\triangleright$ Jaccard similarity matrix
4: $\mathbf{D} \leftarrow 1 - \mathbf{S}$	$\triangleright$ Jaccard distance matrix
5: $\mathbf{W} \leftarrow \mathbf{A} \odot \mathbf{D}$	$\triangleright$ element-wise product between $\mathbf{A}$ and $\mathbf{D}$
6: Compute $\mathbf{s}$ using Equation 3	$\triangleright$ The right dominant eigenvector of $\mathbf{W}$
7: return $\mathbf{s}$	$\triangleright n \times 1$ column vector of non-negative scores

be those that combine a logic of proximity to the most central with *closure* (Coleman 1990), that is, connecting to others who are also linked to their neighbors. This is particularly the case in networks that feature a core-periphery or “nested shell” structure with a well-connected group of nodes that are also mutually connected at the center and another set of less well-connected nodes at the periphery (Borgatti and Everett 2000)—also referred to as the “rich club” phenomenon (Colizza et al. 2006). In this case, eigenvector scoring of the *similarity*-weighted adjacency matrix $\mathbf{S}$ would make more sense and likely yield a substantively different ranking of the nodes than the Alvarez-Socorro et al. (2015) approach based on dissimilarity edge weights.

Given these competing analytic aims, it would seem appropriate to generalize Alvarez-Socorro et al.’s approach to give analysts the flexibility to specify either dissimilarity (brokerage) or similarity (closure) as the preferred criterion for eigenvector-based node ranking, or perhaps attempt to balance between the two.

One option would be to modify Equation 2 as follows:

$$\mathbf{W} = \mathbf{A} \odot [\mathbf{S}^\alpha \odot (1 - \mathbf{S})^\beta] \quad (6)$$

With the restrictions that $0 \leq \alpha \leq 1$, $0 \leq \beta \leq 1$, and $\alpha + \beta \leq 1$.

Note that when $\alpha = \beta = 0$ in Equation 6, then $\mathbf{W} = \mathbf{A}$, and when apply Equation 3 we recover the standard Bonacich eigenvector scores given by Equation 1. When $\alpha = 0$ and $\beta = 1$, then $\mathbf{W} = \mathbf{A} \odot (1 - \mathbf{S})$, which is the same as Equation 2, and when apply Equation 3, we recover the dissimilarity-weighted eigenvector scores of Alvarez-Socorro et al. (2015). When $\alpha = 1$ and $\beta = 0$, then $\mathbf{W} = \mathbf{A} \odot \mathbf{S}$, and when we apply Equation 3, we recover scores that weigh the ranking by the focal node’s *similarity* to well-connected (direct and indirect) neighbors. Finally, when $\alpha > 0$ and $\beta > 0$ then $\mathbf{W}$ is given by the full Equation 6, and when we apply Equation 3, we recover a compromise ranking that takes both brokerage and closure relative to well-connected direct and indirect neighbors into account.

The pseudocode for a revised version of Algorithm 1 that implements this idea is shown in Algorithm 2.

Algorithm 2 Proposed Algorithm for Generalized Similarity Scoring

Require: $\mathbf{A}, \alpha, \beta$ $\triangleright n \times n$ symmetric adjacency matrix and two scalar parameters
Ensure: $0 \leq \alpha \leq 1$
Ensure: $0 \leq \beta \leq 1$
Ensure: $\alpha + \beta \leq 1$

- 1: $\mathbf{P} \leftarrow \mathbf{A}^2$ $\triangleright$ Common neighbors matrix
- 2: $\mathbf{Q} \leftarrow \mathbf{A}(1 - \mathbf{A})$ $\triangleright$ Non-shared neighbors matrix
- 3: $\mathbf{S} \leftarrow \mathbf{P} \div (\mathbf{P} + \mathbf{Q} + \mathbf{Q}^T)$ $\triangleright$ Jaccard similarity matrix
- 4: $\mathbf{D} \leftarrow 1 - \mathbf{S}$ $\triangleright$ Jaccard distance matrix
- 5: $\mathbf{Z} \leftarrow \mathbf{S}^\alpha \odot \mathbf{D}^\beta$ $\triangleright$ Element-wise product of $\mathbf{S}^\alpha$ and $\mathbf{D}^\beta$
- 6: $\mathbf{W} \leftarrow \mathbf{A} \odot \mathbf{Z}$ $\triangleright$ element-wise product between $\mathbf{A}$ and $\mathbf{Z}$
- 7: Compute $\mathbf{s}$ using Equation 3 $\triangleright$ The right dominant eigenvector of $\mathbf{W}$
- 8: **return** $\mathbf{s}$ $\triangleright n \times 1$ column vector of non-negative scores

Data Example

To get a sense of how the generalized eigenvector similarity approach works, I analyze the oft-considered Florentine Families network of Padgett and Ansell (1993)—also analyzed, *inter alia*, by Alvarez-Socorro et al. (2015). More specifically, I used the reduced version of the Florentine Families data analyzed by Breiger and Pattison (1986) available in the R package `networkdata` developed and maintained by Schoch (2022).¹ I construct the network by merging the marriage and business ties and linking two families if they are connected by either or both types of ties. The network can therefore be represented by an undirected graph G and its associated symmetric adjacency matrix $\mathbf{A}$.

A point and line plot of the resulting Florentine Families network is shown in Figure 1, using the standard `igraph` layout algorithm to determine node position (Csárdi and Nepusz 2006). The network shows, as has been observed in myriad previous analyses, the Medici at the center of a “star” subsection of the graph on the left, connected to alters who are not mutually connected. We should expect, therefore, that they would fare well in a feedback central measure, such as that proposed by Alvarez-Socorro et al.’s (2015) in which being connected to well-connected but locally dissimilar partners counts for more (and they show that in their paper).

However, if we were thinking of centrality as better indexed by connections to well-connected others, who are themselves connected to your neighbors, then we should not expect the Medici to fare as well because most of their edges to others are weighted to the dissimilarity end of the scale. Instead, it is the Peruzzi family, at the center of a “wheel” with connected spokes, who should be the most central, since their edges to others are weighted to the similarity end of the scale.

¹Available at <https://schochastics.github.io/networkdata/>.

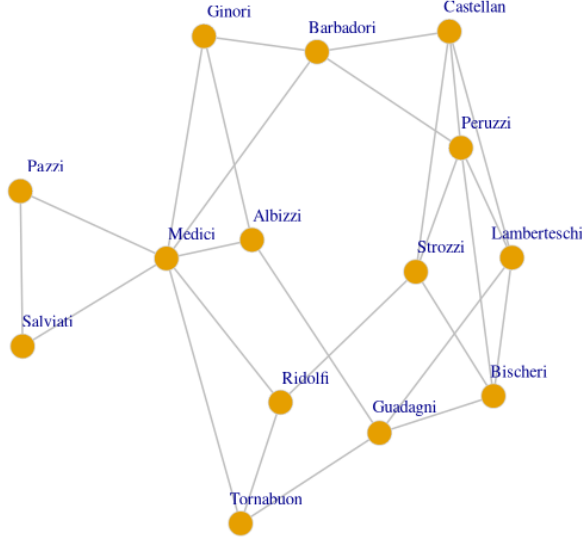


Figure 1: Combined Medici Marriage and Business Network

Results

Table 1 shows the results (all scores are normalized to fall within the zero to one range).² The first two columns of the table show the rankings according to the standard Bonacich (1972) centrality, computed using Equation 6 setting $\alpha = 0$ and $\beta = 0$. Columns three and four show the Alvarez-Socorro et al. (2015) dissimilarity weighted rankings based on scores computed according to Equation 6, setting $\alpha = 0$ and $\beta = 1$. We can see that both the Peruzzi and the Medici families emerge as the most central when using raw eigenvector scores (with the Peruzzi edging out the Medici for the top spot). However, as shown in columns three and four of the table, we can see that once we adopt the Alvarez-Socorro et al. (2015) approach, which rewards both well-connectedness *and* brokerage, the Medici emerge as the undisputed most central actors in the network, consistent with Padgett and Ansell’s (1993) classic analysis, which relied on the betweenness centrality (another metric that rewards brokerage). The Peruzzi, scored as the most central in the standard eigenvector scoring, drop to a distant third place.

The fifth and sixth columns of the table show the rankings obtained from Equation 6, first (columns five and six) using $\alpha = 1$ and $\beta = 0$ (giving all weight to being connected to well-connected others who are also adjacent to your other neighbors). From this

²Replication code to reproduce all of the tables and plots in this paper, as well as an R function implementing Algorithm 2, can be found at <https://github.com/olizardo/combining-similarity-eigenvector>.

Table 1: Eigenvector-Based Scores for the Medici Network.

Node	Eig.	Node	Eig./Dis	Node	Eig./Sim	Node	Eig./Hybrid
Peruzzi	1.000	Medici	1.000	Peruzzi	1.000	Peruzzi	1.000
Medici	0.783	Barbadori	0.641	Castellan	0.752	Barbadori	0.809
Castellan	0.779	Peruzzi	0.556	Lamberteschi	0.679	Castellan	0.794
Lamberteschi	0.744	Guadagni	0.538	Bischeri	0.652	Lamberteschi	0.720
Bischeri	0.729	Strozzi	0.491	Strozzi	0.514	Medici	0.701
Barbadori	0.717	Castellan	0.448	Barbadori	0.259	Bischeri	0.674
Strozzi	0.703	Tornabuon	0.438	Guadagni	0.197	Strozzi	0.581
Guadagni	0.513	Bischeri	0.437	Ginori	0.058	Ginori	0.358
Ginori	0.349	Lamberteschi	0.427	Medici	0.045	Guadagni	0.198
Ridolfi	0.345	Ridolfi	0.427	Albizzi	0.012	Albizzi	0.104
Albizzi	0.308	Albizzi	0.419	Pazzi	0.003	Ridolfi	0.009
Tornabuon	0.307	Ginori	0.348	Salviati	0.003	Tornabuon	0.009
Pazzi	0.000	Pazzi	0.000	Ridolfi	0.000	Pazzi	0.000
Salviati	0.000	Salviati	0.000	Tornabuon	0.000	Salviati	0.000

similarity-based scoring, we can see that the Peruzzi are the undisputed most central actors in the network, while the Medici are fairly peripheral. The other top actors, according to this scoring, all belong to the well-connected core actors revolving around Peruzzi on the right-hand side of Figure 1, which are the closest to a well-connected “core” of central actors in the graph (Borgatti and Everett 2000). The Peruzzi belong to the set of families who, along with the Albizzi, are described as part of the old Patrician “superelite” in Florence in the original work by Padgett and Ansell (1993: 1284). In this respect, it makes sense that they rank highly if we presume a network strategy of closure rather than brokerage.

What if we wanted to combine *both* brokerage *and* closure in determining the eigenvector rankings? Here, it would be useful to have a sense of how the rankings respond to different weights assigned to these two factors when determining the edge weights according to Equation 6. To that end, Figure 2 shows a bump chart plotting the rank order of actors in the Florentine Families network with different combinations of α and β values, exploring the subset of the parameter space that obeys the constraint $\alpha + \beta = 1$. The x-axis in the plot represents different values of β , ranging from zero on the left (and thus $\alpha = 1$) to 1.0 on the right (and thus $\alpha = 0$) by increments of 0.05. The y-axis represents the corresponding eigenvector centrality score computed according to Equation 6. The relative vertical positioning of actors in the plot indicates their rank order according to the computed score, and the connected-line plot gives us a sense of each actor’s rank trajectory in this space. The leftmost ranking in Figure 2 ($\beta = 0$) thus reproduces columns five and six of Table 1 (the similarity weighted rankings), and the rightmost ranking ($\beta = 1$) reproduces columns three and four of the same table (the dissimilarity weighted rankings).

Moving from left to right on the bump chart in Figure 2 thus implies gradually giving more weight to brokerage than to closure when determining the eigenvector scores.

As we can see, the rankings are fairly stable across most values of β , suggesting that similarity weights dominate over the dissimilarity weights in this network when computing the modified eigenvector scores. This is exemplified by the steady dominance of the Peruzzi family across most of the parameter range. However, as we give more weight to dissimilarity weighted edges, the ranks begin to reshuffle—with the Medici family gradually rising up the ranks—toward something resembling the Alvarez-Socorro et al. (2015) scores in the region where $\beta > 0.80$, on the rightmost part of the plot, suggesting that we can determine a compromise ranking that takes into account both similarity and dissimilarity information in the network by drawing parameter values from this area.

Columns seven and eight of Table 1 provide one such snapshot, with $\beta = 0.85$ corresponding to the dashed red vertical line in Figure 2. We can see that the “hybrid” approach, as expected, produces a compromise ranking in which nodes combining both brokerage and similarity-based connectivity strategies appear toward the top. The Barbadori family is notable in this respect: They are a distant second in the approach of Alvarez-Socorro et al. (2015) and in sixth place when ranking nodes according to the similarity weight, but they appear in second place in the hybrid rank, suggesting they do a good job of combining both brokerage and closure connectivity strategies.

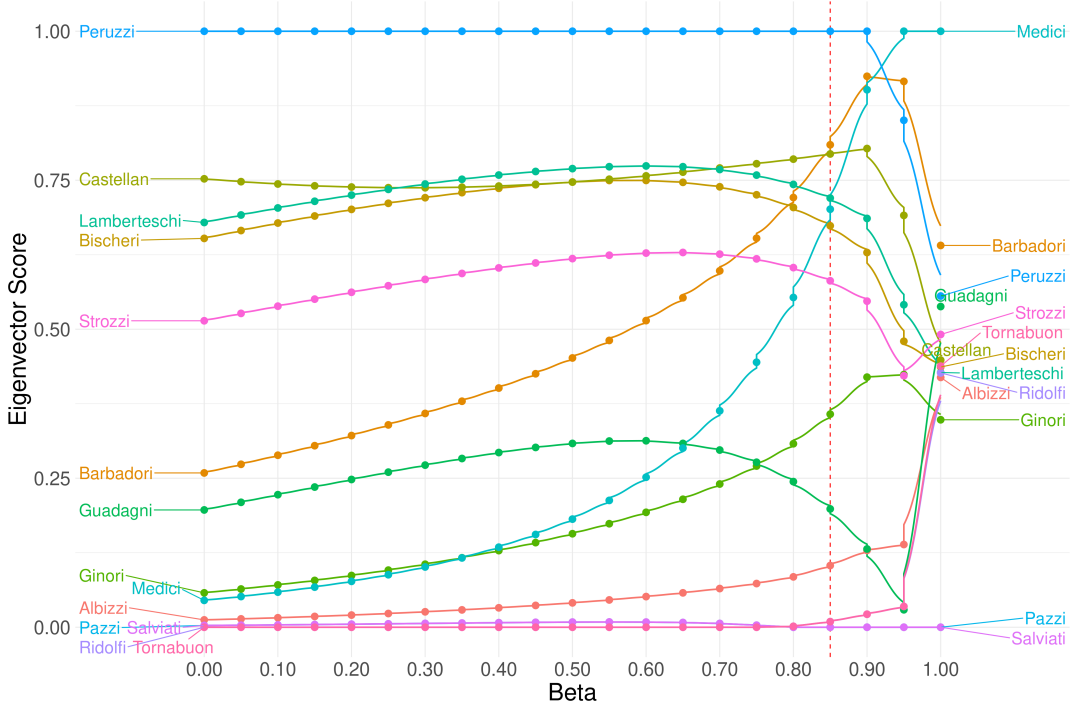


Figure 2: Parameter Sensitivity Analysis for the Medici Network.

Discussion and Concluding Remarks

This short note presents a generalized approach to combining the reasoning behind analyses of social networks that focus on pairwise similarities between nodes based on their local neighborhood (Leicht et al. 2006) with spectral ranking based on the idea of feedback centralities, where nodes are central when they are directly and indirectly connected to central others (Bonacich 1972).

Building on recent work by Alvarez-Socorro et al. (2015) who propose to use the pairwise dissimilarities between nodes as a weight on each edge before estimating the usual Bonacich eigenvector rankings, I argue that in some settings it might be more desirable to weigh the edges by the *similarity*, as this may be more consistent with the idea of prominence as membership in a centralized core of well-connected nodes. Alternatively, the analyst may want to consider both dissimilarities and similarities as edge weights in computing the ranking, rewarding actors who engage in hybrid brokerage and closure strategies the most. In this way, the generalized approach proposed here allows us to *combine* these two analytic strategies, ranking nodes highly when they hedge their bets and engage in both connectivity strategies.

Analysis of the classic Florentine Families dataset of Padgett and Ansell (1993) reveals that, when considering two different connectivity strategies, the much heralded Medici family emerges as *brokerage specialists*: Undisputedly the top nodes when the pairwise dissimilarities between connected dyads are used as edge weights before eigenvector ranking (Alvarez-Socorro et al. 2015). When we use ranking approaches that reward belonging to well-connected cores, however, the Medici emerge as largely peripheral nodes in the network, with the “superelite” old-moneyed Peruzzi’s emerging as the top node in the combined marriage and financial trade network.

The approach proposed here can also be extended to other families of feedback-based centralities. For instance, the famous all-paths centrality due to Katz (1953) could be extended in the same way, by substituting $\mathbf{W}$ for $\mathbf{A}$ in the usual calculation. Here, we would consider them central if they share many paths with many others, with the paths weighted by similarity or dissimilarity, depending on the application. The same approach can be used with other attenuated path-based measures of node position, such as communicability centrality (Estrada and Higham 2010), which can also be fruitfully combined with similarity/dissimilarity analysis.

References

- Alvarez-Socorro, A. J., G. C. Herrera-Almarza, and L. A. González-Díaz. 2015. “Eigen-centrality based on dissimilarity measures reveals central nodes in complex networks.” *Scientific Reports* 5:17095.
- Bonacich, Phillip. 1972. “Factoring and weighting approaches to status scores and clique identification.” *The Journal of Mathematical Sociology* 2:113–120.
- Bonacich, Phillip. 1987. “Power and Centrality: A Family of Measures.” *American Journal of Sociology* 92:1170–1182.
- Borgatti, Stephen P. 2005. “Centrality and Network Flow.” *Social networks* 27:55–71.
- Borgatti, Stephen P and Martin G Everett. 2000. “Models of core/periphery structures.” *Social Networks* 21:375–395.
- Borgatti, Stephen P and Martin G Everett. 2006. “A graph-theoretic perspective on centrality.” *Social networks* 28:466–484.
- Breiger, Ronald L and Philippa E Pattison. 1986. “Cumulated Social Roles: The Duality of Persons and their Algebras.” *Social networks* 8:215–256.
- Burt, Ronald S. 1995. *Structural Holes*. Harvard University Press.
- Coleman, James S. 1990. *Foundations of Social Theory*. Cambridge, MA: Harvard University Press.
- Colizza, Vittoria, Alessandro Flammini, M Angeles Serrano, and Alessandro Vespignani. 2006. “Detecting rich-club ordering in complex networks.” *Nature physics* 2:110–115.
- Csárdi, Gábor and Tamás Nepusz. 2006. “The igraph software package for complex network research.” *InterJournal Complex Systems*:1695.
- Estrada, Ernesto and Desmond J. Higham. 2010. “Network Properties Revealed through Matrix Functions.” *SIAM Review* 52:696–714.
- Fouss, Francois, Alain Pirotte, and Marco Saerens. 2005. “A novel way of computing similarities between nodes of a graph, with application to collaborative recommendation.” *The 2005 IEEE/WIC/ACM International Conference on Web Intelligence (WI’05)* pp. 1–7.
- Katz, Leo. 1953. “A new status index derived from sociometric analysis.” *Psychometrika* 18:39–43.
- Leicht, E. A., Petter Holme, and M. E. J. Newman. 2006. “Vertex similarity in networks.” *Physical Review E* 73:026120.

- Padgett, John F and Christopher K Ansell. 1993. “Robust Action and the Rise of the Medici, 1400-1434.” *The American journal of sociology* 98:1259–1319. ID: 276.
- Schoch, David. 2022. “networkdata: Repository of Network Datasets (v0.1.14).” <https://doi.org/10.5281/zenodo.7189928>.
- Vigna, Sebastiano. 2016. “Spectral ranking.” *Network Science* 4:433–445.